

Natural Language Processing and Neural Networks

Steps of NLP

- Natural Language Processing involves a sequence of tasks that convert raw text into structured, machine-understandable formats:
 - Natural language processing (NLP) is a subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate with human language.
1. Tokenization: Splitting text into words or tokens.
 2. Stop-word Removal: Eliminating common words like 'the', 'is' that add little meaning.
 3. Stemming/Lemmatization: Reducing words to their root forms.
 4. POS Tagging: Labeling each word with its grammatical role.
 5. Parsing: Analyzing sentence structure.
 6. Semantic Analysis: Understanding meanings and relationships.
 7. Discourse and Pragmatic Analysis: Interpreting context and intent.

Applications of NLP

- NLP powers a wide array of modern tools:
 - Chatbots and virtual assistants like Siri or Alexa.
 - Text summarization tools that condense articles.
 - Sentiment analysis to gauge opinions on social media.
 - Language translation services.
 - Spam detection and content moderation.
 - Document categorization and search engines.



Importance of NLP

- NLP is crucial for enabling machines to understand human languages:
 - Makes human-computer interaction more natural.
 - Facilitates intelligent systems in various domains.
 - Automates labor-intensive language tasks.
 - Enables accessibility for diverse populations through speech-to-text tools.

Limitations of NLP

- Despite progress, NLP faces challenges:
 - Ambiguity in language and multiple interpretations.
 - Difficulty understanding sarcasm or idioms.
 - High dependence on context for accurate translation.
 - Complexities of multilingual and domain-specific language.
 - Requires large datasets and computational power.



Ambiguity

- Words and sentences can have multiple meanings



Sarcasm

- Implicit meaning opposite of literal meaning

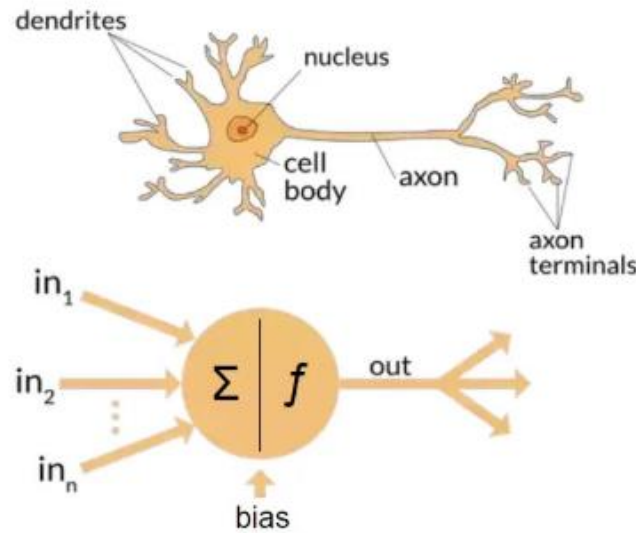


Multilingual

- Diversity of languages with different syntax

Biological vs Artificial Neural Network

- Biological neurons in the brain consist of dendrites, a cell body, and an axon. They process signals and adapt via synaptic strength.
- Artificial Neural Networks mimic this structure:
 - Neurons (nodes) process input and produce output using activation functions.
 - Weighted connections adjust during training to 'learn' patterns.



Applications of Neural Networks

- Neural networks are foundational to many AI advancements:
 - Image and speech recognition systems (e.g., face unlock, transcription).
 - Autonomous vehicles using object detection and navigation.
 - Financial models for fraud detection or stock prediction.
 - Medical diagnostics like tumor detection from scans.

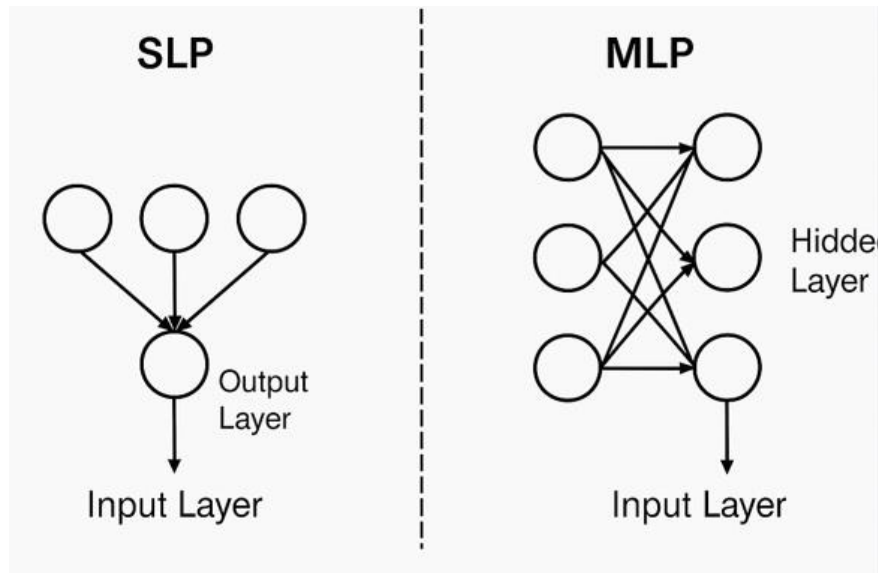


Perceptron Learning Algorithm

- The Perceptron is the simplest neural network model. It works by:
 1. Initializing weights randomly.
 2. Calculating the output using an activation function.
 3. Comparing output with actual result and computing error.
 4. Updating weights based on the error.
 5. Repeating until output is accurate enough.

SLP vs MLP

- Single Layer Perceptron (SLP):
 - Only input and output layers.
 - Solves linearly separable problems.
- Multi-Layer Perceptron (MLP):
 - Has hidden layers.
 - Can solve non-linear, complex problems.
 - Uses non-linear activation functions like ReLU, sigmoid.



Backpropagation Algorithm

- Backpropagation is a supervised learning algorithm used in MLPs:
 1. A forward pass calculates the prediction.
 2. Loss is computed using a loss function.
 3. Error is propagated backward using gradients.
 4. Weights are updated using optimization algorithms like gradient descent.
- This process continues until minimal error is achieved.

